

Simultaneous Source Localization and Formation via a Distributed Sign Gradient-Free Algorithm

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Abstract—This article develops a distributed sign gradient-free algorithm for simultaneous source localization and formation of a multirobot system. A distinguished feature of the algorithm is that it takes into account robots' measure noise as well as ternary communication, which significantly reduces the communication cost of the overall network. Considering the presence of noise, the algorithm is designed to be gradient-free. In addition, an inherent connection is established between the selection of control parameters and the convergence property of the sign gradient-free algorithm, as well as signal strength, noisy intensity, formation radius, and the number of informed robots.

Index Terms—Distributed source localization, formation control, gradient-free algorithm, multiagent system, signum function.

I. INTRODUCTION

RECENTLY, there has been considerable attention devoted to the study of source localization via mobile robotic networks [1], [2]. Tasks of this kind arise from gas leakage detection, sound source localization, environmental monitoring, and search-and-rescue operations in disaster areas, to name just a few [3], [4], [5]. As information transmission is inevitable and is generally noisy and costly, it may hamper the attainable performance of the robotic networks. Today, it is well

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known that communication network topologies and information transmission noise both pose significant challenges that are unseen previously for the tasks of distributed source localization [6], [7], [8]. This requires the re-examination of stability and control issues for cooperative multirobot source localization.

Recent studies of single-robot source localization have developed several effective algorithms, such as information-based search [9], stochastic source seeking [10], and fluid-simulation-based approaches [11]. These algorithms are easy to implement, yet the performance remains unsatisfactory because the task might fail when the searching environment is large and complex. Despite the significant progress on single-robot source localization, the issues for cooperative multirobot source localization remain largely unsolved. Toward this end, gradient-based algorithms are proposed for a group of robots to achieve source localization assuming that the gradient information is accessible; see [12] and the references therein. In this case, the additional requirement on robot formation might be imposed, leading to the simultaneous source localization and formation problem [13], [14]. It is worth pointing out that in practice, the measurements of robots cannot be perfect, where, for instance, the gradients of the measurements are unaccessible or inaccurate [15]. In [16], a strategy is proposed to estimate the gradient and Hessian of robots' measurements through symmetric formations with a reduced number of robots for source seeking. Different from the cooperative control method with full communication in [16], the works in [8], [17], and [18] offer cooperative source location algorithms for multiple agents with communication constraints. In [18], a least-squared gradient estimation strategy is proposed. Meanwhile, in the absence of gradient information, an extremum-seeking strategy is suggested [19], [20]. In particular, in [19], a global extremum-seeking method is proposed using noise-corrupted measurements, yet the method is only applicable to the scenario where the source is outside the admissible area. The results were extended to the case of double-integrator systems even under deception attacks in [21].

This article addresses the issue of simultaneous source localization and formation control for a cooperative multirobot network with noisy measurements and ternary communication. An algorithm will be developed, which offers a number of appealing features. First, it takes into account noisy measurements, which are corrupted by bounded additive noise, with limited ranges of

robot sensors; thus it is more practical for applications. Second, it employs the signum function to encode neighboring information. Consequently, a pair of neighboring information can have only three values: 0, 1, and -1 , i.e., the neighboring information is ternary. This reduces to a large extent the computation and communication costs, and renders the system more suitable for large-scale networks. Although the introduction of the signum function makes the convergence analysis significantly more challenging, it will be demonstrated that as long as the formation gain is sufficiently large, it is plausible to reach simultaneous source localization and formation. Meanwhile, it will show how the signal strength, noisy intensity, formation radius, and number of informed robots might crucially affect the localization error, thus shedding some new insights into the intrinsic mechanism of multirobot cooperative tasks.

The rest of this article is organized as follows. Section II introduces mathematical preliminaries and formulates the problem. Section III describes the control design as well as the convergence analysis. Section IV shows a numerical example to demonstrate the effectiveness of the proposed algorithm. Section V presents the experimental results via a group of TurtleBot3 robots. Finally, Section VI concludes this article.

II. PRELIMINARIES AND PROBLEM INSTALLATION

A. Notation

Let $\mathbb{R}$, $\mathbb{R}^n$, and $\mathbb{R}^{m \times n}$ denote, respectively, the sets of real numbers, n -dimensional real vectors, and $m \times n$ real matrices. For a symmetric matrix A , let $\lambda_{\min}(A)$ and $\lambda_{\max}(A)$ denote, respectively, the smallest and largest eigenvalues of A . Let $\|\cdot\|$ denote the Euclidian norm for vectors and the induced two-norm for matrices. Let $\text{sgn}(\cdot)$ be the signum function. For a function $f(\cdot)$, let $\nabla f(\cdot)$ and $H_f(\cdot)$ denote, respectively, its gradient and Hessian matrix.

B. Graph Theory

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be an undirected graph modeling interactions among robots, where $\mathcal{V} = \{1, \dots, n\}$ is the set of robots and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the set of edges. An edge $(i, j) \in \mathcal{E}$ indicates that robots i and j could exchange information. Let $\mathcal{N}_i = \{i \in \mathcal{V} : a_{ij} \neq 0\}$ denote the set of neighbors of robot i . Let $\mathcal{A} = [a_{ij}] \in \mathbb{R}^{n \times n}$ denote the adjacency matrix of the graph $\mathcal{G}$, where $a_{ij} = 1$ if $(i, j) \in \mathcal{E}$ and $a_{ij} = 0$ otherwise. Throughout this article, the following assumption is made.

Assumption 1: The graph $\mathcal{G}$ is undirected and connected.

C. Problem Installation

Consider a group of n mobile robots moving in the $\mathbb{R}^2$ space. The model of robot i is given by

$$\dot{p}_i(t) = u_i(t), \quad i \in \mathcal{V} \quad (1)$$

where $p_i(t) \in \mathbb{R}^2$ and $u_i(t) \in \mathbb{R}^2$ denote, respectively, the position and the control input of robot i . Let $p_s \in \mathbb{R}^2$ denote the position of a source, which is capable of emitting a certain signal. Assume that all the signals, such as light, radiation, and sound signals, satisfy the inverse-square law. In this situation, the intensity of linear waves radiating from a point source is

inversely proportional to the square of the distance from the source. Specifically, let $f(z) = \kappa \|z - p_s\|^{-2} : \mathbb{R}^2 \rightarrow \mathbb{R}^+$, $z \in \mathbb{R}^2$ be the signal distribution function, which attains the unique minimum value at $z = p_s$ with $\kappa > 0$ being a constant determined by the properties of the signals (see also [8] and [22]).

The measurement of robot i is given by the following function:

$$\sigma(p_i) = \begin{cases} f(p_i) + \eta(p_i), & \text{if } \|p_i - p_s\| < D_{\max} \\ f_{D_{\max}}, & \text{otherwise} \end{cases} \quad (2)$$

where $D_{\max}$ denotes the maximum sensing distance of the robots, $f_{D_{\max}} = \kappa D_{\max}^{-2} + \delta$ the maximum sensing value of the robots, and $\eta(p_i)$ the perturbation that may come from the sensors or uncertainties of the field, with $\delta > 0$ being a small constant. For further analysis, let $\mathcal{X} = \{i \in \mathcal{V} : \|p_i - p_s\| < D_{\max}\}$ be the set of informed robots, i.e., robots capable of accessing the signal of the source. In addition let $\mathcal{Y} = \mathcal{V} \setminus \mathcal{X}$, and let $n_{\mathcal{X}}$ and $n_{\mathcal{Y}}$ denote, respectively, the number of robots in $\mathcal{X}$ and $\mathcal{Y}$. Apparently, $n_{\mathcal{X}} + n_{\mathcal{Y}} = n$. The following assumption is made on the perturbation function.

Assumption 2: The perturbation function $|\eta(p_i)| \leq \delta$ for some $\delta > 0$, and there exists at least one robot p_i satisfying $\|p_i - p_s\| < D_{\max}$.

Remark 1: Assumption 2 indicates that the perturbation is upper bounded, and there exists at least one robot that can access the signal emitted from the source.

The objective is to design a distributed control input $u_i(t)$ for system (1) by employing only local information such that the group of robots forms a circular shape, and the formation center $p^* = \frac{1}{n} \sum_{i=1}^n p_i$ is arbitrarily close to the source p_s despite the presence of perturbation. Mathematically, the objective is described by

$$\begin{aligned} \lim_{t \rightarrow \infty} p_i(t) - R\phi(\theta_i) &= p^*(t), \text{ for all } i \in \mathcal{V} \\ \lim_{t \rightarrow \infty} \|p^*(t) - p_s\| &< \epsilon \end{aligned} \quad (3)$$

where $\phi(\theta_i) = [\cos \theta_i, \sin \theta_i]^T$, $\theta_i = 2\pi i/n$, $0 < R \leq D_{\max}$ with R being the radius of circular formation and ϵ an arbitrarily small constant.

III. CONTROL DESIGN AND CONVERGENCE ANALYSIS

The distributed control law of robot i is designed as follows:

$$\begin{aligned} u_i(t) &= \alpha \sum_{j \in \mathcal{N}_i} \text{sgn}((p_j(t) - R\phi(\theta_j)) - (p_i(t) - R\phi(\theta_i))) \\ &\quad - \frac{2\beta}{R} \sigma(p_i(t)) \phi(\theta_i) \end{aligned} \quad (4)$$

where α and $\beta \in \mathbb{R}^+$ are the two gain parameters, and $\text{sgn}(\cdot)$ is the signum function. In (4), the first term on the right-hand side is used for formation control, while the second term is for source localization. Due to the presence of disturbance, the signal $f(p_i)$ cannot be directly measured. Consequently, its gradient is unknown. The second term is, hence, employed as an approximation of the gradient, which relies only on $\sigma(p_i)$ and $\phi(\theta_i)$.

Remark 2: One benefit of employing the signum function in (4) is that the cost for exchanging information is reduced

largely, as only two bits are needed to encode the neighboring information.

Remark 3: Note that rigidity-relevant assumptions are usually imposed on distance-based formation control, yet the sign gradient-free algorithm (4) is based on the displacement information. Therefore, the rigidity assumptions are not needed in the present work.

The main theoretical results are summarized in the following theorem.

Theorem 1: Let Assumptions 1 and 2 hold. For system (1) with the control law (4), if $\frac{\alpha}{\beta} > \frac{4nf_{D\max}}{R}$, the objective (3) is achieved asymptotically with

$$\epsilon = \frac{2\pi n\delta}{\kappa R(2\pi n_{\mathcal{X}} - n|\sin(\frac{2\pi n_{\mathcal{X}}}{n})|)}.$$

Proof: It is first shown that a circular formation is reached in finite time. Let $z_i(t) = p_i(t) - R\phi(\theta_i)$ and $z^*(t) = \frac{1}{n} \sum_{j=1}^n z_j(t)$. It follows that $z^*(t) = \frac{1}{n} \sum_{j=1}^n z_j(t) = \frac{1}{n} \sum_{j=1}^n p_j(t) - \frac{R}{n} \sum_{j=1}^n \phi(\theta_j) = p^*(t)$. The derivative of z_i is given by $\dot{z}_i(t) = \dot{p}_i(t) = u_i(t)$. Note that

$$\|\sigma(p_i(t))\phi(\theta_i)\| = \sigma(p_i(t)) \leq f_{D\max}$$

for all $i \in \mathcal{V}$. Define the Lyapunov function candidate for circular formation

$$V_c(t) = \frac{1}{2} \sum_{i=1}^n \|z_i(t) - z^*(t)\|^2. \quad (5)$$

It is noted that $V_c(t)$ has value 0 if $z_i(t) = z^*(t)$ for all $i = 1, \dots, n$, and $V_c(t) > 0$ otherwise. The derivative of $V_c(t)$ is given by

$$\begin{aligned} \dot{V}_c(t) &= \sum_{i=1}^n (z_i(t) - z^*(t))^T \\ &\times \left(\alpha \sum_{j \in \mathcal{N}_i} \text{sgn}(z_j(t) - z_i(t)) - \frac{2\beta}{R} \sigma(p_i(t)) \phi(\theta_i) \right) \\ &- \sum_{i=1}^n (z_i(t) - z^*(t))^T \dot{z}^*(t) \\ &= \alpha \sum_{i=1}^n (z_i(t) - z^*(t))^T \sum_{j \in \mathcal{N}_i} \text{sgn}(z_j(t) - z_i(t)) \\ &- \frac{2\beta}{R} \sum_{i=1}^n (z_i(t) - z^*(t))^T \sigma(p_i(t)) \phi(\theta_i). \quad (6) \end{aligned}$$

Since the graph $\mathcal{G}$ is undirected and $\|\sigma(p_i(t))\phi(\theta_i)\| \leq f_{D\max}$, it follows from (6) that:

$$\begin{aligned} \dot{V}_c(t) &= \frac{\alpha}{2} \sum_{i=1}^n \sum_{j \in \mathcal{N}_i} (z_i(t) - z_j(t))^T \text{sgn}(z_j(t) - z_i(t)) \\ &- \frac{2\beta}{R} \sum_{i=1}^n (z_i(t) - z^*(t))^T \sigma(p_i(t)) \phi(\theta_i) \\ &\leq \frac{\alpha}{2} \sum_{i=1}^n \sum_{j \in \mathcal{N}_i} (z_i(t) - z_j(t))^T \text{sgn}(z_j(t) - z_i(t)) \end{aligned}$$

$$+ \frac{2\beta f_{D\max}}{R} \sum_{i=1}^n \|z_i(t) - z^*(t)\|. \quad (7)$$

Next, consider the formation control error $z_i(t) - z^*(t)$ for all $i \in \mathcal{V}$. Let $i_0, j_0 \in \mathcal{V}$ be the integers such that $\|z_{i_0}(t) - z_{j_0}(t)\| = \max_{i,j \in \mathcal{V}} \|z_i(t) - z_j(t)\|$ at time t . It follows that $\|z_i(t) - z^*(t)\| \leq \frac{1}{n} \sum_{j=1}^n \|z_i(t) - z_j(t)\| \leq \|z_{i_0}(t) - z_{j_0}(t)\|$. Since the graph $\mathcal{G}$ is connected, there must exist a path $(i_0, i^0), (i^0, i^1), \dots, (i^{h-1}, i^h), (i^h, j_0)$ connecting robots i_0 and j_0 . Note that $\|z_{i_0}(t) - z_{j_0}(t)\| \leq \|z_{i_0}(t) - z_{i^0}(t)\| + \sum_{k=1}^h \|z_{i^{k-1}}(t) - z_{i^k}(t)\| + \|z_{i^h}(t) - z_{j_0}(t)\| \leq \sum_{i=1}^n \sum_{j \in \mathcal{N}_i} \|z_i(t) - z_j(t)\|$, leading to $\|z_{i_0}(t) - z_{j_0}(t)\| \leq \sum_{i=1}^n \sum_{j \in \mathcal{N}_i} \|z_i(t) - z_j(t)\|$. Meanwhile, one has $\|z_i(t) - z_j(t)\| \leq -[z_i(t) - z_j(t)]^T \text{sgn}(z_j(t) - z_i(t))$ for all $i, j \in \mathcal{V}$. It thus follows that:

$$\begin{aligned} \|z_i(t) - z^*(t)\| &\leq \|z_{i_0}(t) - z_{j_0}(t)\| \\ &\leq - \sum_{i=1}^n \sum_{j \in \mathcal{N}_i} (z_i(t) - z_j(t))^T \text{sgn}(z_j(t) - z_i(t)). \quad (8) \end{aligned}$$

If $\alpha/\beta > 4nf_{D\max}/R$, then

$$\begin{aligned} \dot{V}_c(t) &\leq \frac{\alpha}{2} \sum_{i=1}^n \sum_{j \in \mathcal{N}_i} (z_i(t) - z_j(t))^T \text{sgn}(z_j(t) - z_i(t)) \\ &- \frac{2n\beta f_{D\max}}{R} \sum_{i=1}^n \sum_{j \in \mathcal{N}_i} (z_i(t) - z_j(t))^T \\ &\times \text{sgn}(z_j(t) - z_i(t)) \\ &= \sum_{i=1}^n \sum_{j \in \mathcal{N}_i} (z_i(t) - z_j(t))^T \\ &\times \text{sgn}(z_j(t) - z_i(t)) \left(\frac{\alpha}{2} - \frac{2n\beta f_{D\max}}{R} \right) \leq 0. \end{aligned}$$

This implies that $V_c(t)$ and $\|z_{i_0}(t) - z_{j_0}(t)\|$ are bounded. When $V_c(t) \neq 0$, i.e., $\|z_{i_0}(t) - z_{j_0}(t)\| \neq 0$, it follows from (8) that $\sqrt{V_c(t)} \leq \sqrt{\frac{n}{2}} \|z_{i_0}(t) - z_{j_0}(t)\|$, which yields:

$$\begin{aligned} \frac{\dot{V}_c(t)}{\sqrt{V_c(t)}} &\leq \frac{\sum_{i=1}^n \sum_{j \in \mathcal{N}_i} (z_i(t) - z_j(t))^T \text{sgn}(z_j(t) - z_i(t))}{\frac{1}{\sqrt{2}} \sqrt{n} \|z_{i_0}(t) - z_{j_0}(t)\|} \\ &\times \left(\frac{\alpha}{2} - \frac{2n\beta f_{D\max}}{R} \right) \\ &\leq - \frac{\sqrt{2n}}{n} \left(\frac{\alpha}{2} - \frac{2n\beta f_{D\max}}{R} \right) \\ &< 0. \end{aligned}$$

Integrating both sides of the above inequality gives

$$2\sqrt{V_c(t)} - 2\sqrt{V_c(0)} < - \frac{\sqrt{2n}}{n} \left(\frac{\alpha}{2} - \frac{2n\beta f_{D\max}}{R} \right) t$$

indicating that $V_c(t)$ will converge to zero in finite time. That is, $z_i(t)$ will reach a consensus at $z^*(t) = p^*(t)$ in a finite time T_c and $p_i(t) - p_j(t) = R\phi(\theta_i) - R\phi(\theta_j)$ when $t \geq T_c$. In other

words, the group of robots will form and maintain the circular formation with the center $p^*(t)$ after $t \geq T_c$.

Next, it is shown that the formation center will converge to the source location p_s . Since the graph $\mathcal{G}$ is undirected and connected, it follows that, when $t \geq T_c$, $z_i(t) = z_j(t) = z^*(t) = p^*(t)$ and:

$$\begin{aligned} \dot{p}^*(t) &= \frac{1}{n} \sum_{i=1}^n \dot{z}_i(t) \\ &= \frac{1}{n} \sum_{i=1}^n \left(\alpha \sum_{j \in \mathcal{N}_i} \text{sgn}(z_j(t) - z_i(t)) - \frac{2\beta}{R} \sigma(p_i(t)) \phi(\theta_i) \right) \\ &= -\frac{2\beta}{nR} \sum_{i=1}^n \sigma(p_i(t)) \phi(\theta_i) \\ &= -\frac{2\beta}{nR} \sum_{i \in \mathcal{X}} (f(p_i(t)) + \eta(p_i(t))) \phi(\theta_i) \\ &\quad - \frac{2\beta}{nR} \sum_{i \in \mathcal{Y}} f_{D\max} \phi(\theta_i). \end{aligned} \quad (9)$$

Let $F_Y = \frac{2\beta}{nR} \sum_{i \in \mathcal{Y}} (f(p_i(t)) + \eta(p_i(t))) \phi(\theta_i)$. It follows from $n_{\mathcal{X}} + n_{\mathcal{Y}} = n$ that:

$$\begin{aligned} \dot{p}^*(t) &= -\frac{2\beta}{nR} \sum_{i \in \mathcal{X}} (f(p_i(t)) + \eta(p_i(t))) \phi(\theta_i) \\ &\quad - F_Y + F_Y - \frac{2\beta}{nR} \sum_{i \in \mathcal{Y}} f_{D\max} \phi(\theta_i) \\ &= -\frac{2\beta}{nR} \sum_{i=1}^n (f(p_i(t)) + \eta(p_i(t))) \phi(\theta_i) \\ &\quad - \frac{2\beta}{nR} \sum_{i \in \mathcal{Y}} (f_{D\max} - (f(p_i(t)) + \eta(p_i(t)))) \phi(\theta_i) \\ &= -\frac{2\beta}{nR} \sum_{i=1}^n f(p_i(t)) \phi(\theta_i) \\ &\quad - \frac{2\beta}{nR} \sum_{i \in \mathcal{X}} \eta(p_i(t)) \phi(\theta_i) \\ &\quad + \frac{2\beta}{nR} \sum_{i \in \mathcal{Y}} (f(p_i(t)) - f_{D\max}) \phi(\theta_i). \end{aligned} \quad (10)$$

Use the first-order Taylor expansion of $f(p_i(t))$ at the point $z_i(t)$ and recall that $p_i(t) - z_i(t) = R\phi(\theta_i)$. Since $z_i(t) = p^*(t)$, the following holds for all $i = 1, \dots, n$ after $t \geq T_c$:

$$\begin{aligned} f(p_i(t)) - f(p^*(t)) &= \nabla f(p^*(t))^T R\phi(\theta_i) \\ &\quad + \frac{1}{2} R\phi(\theta_i)^T H_f(p^*(t)) R\phi(\theta_i). \end{aligned} \quad (11)$$

Multiplying (11) by $(-2\beta/nR)\phi(\theta_i)$ and summing over $i = 1, \dots, n$ lead to

$$-\frac{2\beta}{nR} \sum_{i=1}^n f(p_i(t)) \phi(\theta_i) + f(p^*(t)) \frac{2\beta}{nR} \sum_{i=1}^n \phi(\theta_i)$$

$$\begin{aligned} &= -\frac{2\beta}{nR} \sum_{i=1}^n \nabla f(p^*(t))^T R\phi(\theta_i) \phi(\theta_i) \\ &\quad - \frac{\beta}{nR} \sum_{i=1}^n R\phi(\theta_i)^T H_f(p^*(t)) R\phi(\theta_i) \phi(\theta_i). \end{aligned} \quad (12)$$

Since $\phi(\theta_i) = [\cos \theta_i, \sin \theta_i]^T$ and $\theta_i = 2\pi i/n$, it follows that $\sum_{i=1}^n \phi(\theta_i) = 0$, which yields:

$$\begin{aligned} &= -\frac{2\beta}{nR} \sum_{i=1}^n f(p_i(t)) \phi(\theta_i) \\ &= -\frac{2\beta}{nR} \sum_{i=1}^n \nabla f(p^*(t))^T R\phi(\theta_i) \phi(\theta_i) \\ &\quad - \frac{\beta}{nR} \sum_{i=1}^n R\phi(\theta_i)^T H_f(p^*(t)) R\phi(\theta_i) \phi(\theta_i). \end{aligned} \quad (13)$$

In the following, the second term of (13) is analyzed. Since $\cos^2 \theta_i = 0.5(1 + \cos(2\theta_i))$, $\sin^2 \theta_i = 0.5(1 - \cos(2\theta_i))$, and $\sum_{i=1}^n \cos(2\theta_i) = \sum_{i=1}^n \sin(2\theta_i) = 0$ for $n > 2$, one has

$$\begin{aligned} &= -\frac{2\beta}{nR} \sum_{i=1}^n \nabla f(p^*(t))^T R\phi(\theta_i) \phi(\theta_i) \\ &= -\frac{2\beta}{n} \sum_{i=1}^n \phi(\theta_i) \phi(\theta_i)^T \nabla f(p^*(t)) \\ &= -\frac{2\beta}{n} \left(\sum_{i=1}^n \begin{bmatrix} \cos^2(\theta_i) & 0.5 \sin(2\theta_i) \\ 0.5 \sin(2\theta_i) & \sin^2(\theta_i) \end{bmatrix} \right) \\ &\quad \times \nabla f(p^*(t)) \\ &= -\frac{\beta}{n} \left(\sum_{i=1}^n \begin{bmatrix} (1 + \cos(2\theta_i)) & \sin(2\theta_i) \\ \sin(2\theta_i) & (1 - \cos(2\theta_i)) \end{bmatrix} \right) \\ &\quad \times \nabla f(p^*(t)) \\ &= -\beta \nabla f(p^*(t)). \end{aligned} \quad (14)$$

From trigonometric properties, it follows that:

$$\begin{aligned} \sum_{i=1}^n a_i &= (3/4) \sum_{i=1}^n \cos \theta_i + (1/4) \sum_{i=1}^n \cos 3\theta_i = 0 \\ \sum_{i=1}^n d_i &= (3/4) \sum_{i=1}^n \sin \theta_i - (1/4) \sum_{i=1}^n \sin 3\theta_i = 0 \\ \sum_{i=1}^n b_i &= \sum_{i=1}^n \sin \theta_i - \sum_{i=1}^n d_i = 0 \\ \sum_{i=1}^n c_i &= \sum_{i=1}^n \cos \theta_i - \sum_{i=1}^n a_i = 0. \end{aligned}$$

Letting $B_i = \phi(\theta_i)^T H_f(p^*(t)) \phi(\theta_i) \phi(\theta_i)$ leads to

$$B_i = \begin{bmatrix} a_i H_{f11}(p^*) + b_i (H_{f12}(p^*) + H_{f21}(p^*)) + c_i H_{f22}(p^*) \\ b_i H_{f11}(p^*) + c_i (H_{f12}(p^*) + H_{f21}(p^*)) + d_i H_{f22}(p^*) \end{bmatrix}$$

and $\sum_{i=1}^n B_i = 0$, which yields

$$\begin{aligned} & -\frac{\beta}{nR} \sum_{i=1}^n R\phi(\theta_i)^T H_f(p^*(t)) R\phi(\theta_i)\phi(\theta_i) \\ & = -\frac{\beta R}{n} \sum_{i=1}^n B_i = 0. \end{aligned} \quad (15)$$

It follows from (13) to (15) that:

$$-\frac{2\beta}{nR} \sum_{i=1}^n f(p_i(t))\phi(\theta_i) = -\beta \nabla f(p^*(t)). \quad (16)$$

Substituting (16) into (10) gives

$$\begin{aligned} \dot{p}^*(t) &= -\frac{2\beta}{nR} \sum_{i=1}^n \sigma(p_i(t))\phi(\theta_i) \\ &= -\beta \nabla f(p^*(t)) \\ &\quad -\frac{2\beta}{nR} \sum_{i \in \mathcal{X}} \eta(p_i(t))\phi(\theta_i) \\ &\quad +\frac{2\beta}{nR} \sum_{i \in \mathcal{Y}} (f(p_i(t)) - f_{D_{\max}})\phi(\theta_i). \end{aligned} \quad (17)$$

Define the Lyapunov function candidate for localization as

$$V(p^*(t)) = \frac{1}{2} \|p^*(t) - p_s\|^2.$$

The derivative is given by

$$\begin{aligned} \dot{V}(p^*(t)) &= (p^*(t) - p_s)^T \dot{p}^*(t) \\ &= -\beta (p^*(t) - p_s)^T \nabla f(p^*(t)) \\ &\quad -\frac{2\beta}{nR} (p^*(t) - p_s)^T \sum_{i \in \mathcal{X}} \eta(p_i(t))\phi(\theta_i) \\ &\quad +\frac{2\beta\kappa}{nR} \sum_{i \in \mathcal{Y}} \left(\|p_i - p_s\|^2 - D_{\max}^2 \right) \phi(\theta_i) \\ &\quad -\frac{2\beta}{nR} (p^*(t) - p_s)^T \sum_{i \in \mathcal{Y}} \delta\phi(\theta_i) \end{aligned} \quad (18)$$

for $t > T_c$. Since $|\eta(p_i(t))| \leq \delta$, one has

$$\begin{aligned} & -\frac{2\beta}{nR} (p^*(t) - p_s)^T \sum_{i \in \mathcal{X}} \eta(p_i(t))\phi(\theta_i) \\ & -\frac{2\beta}{nR} (p^*(t) - p_s)^T \sum_{i \in \mathcal{Y}} \delta\phi(\theta_i) \\ & \leq \frac{2\beta}{nR} \|p^*(t) - p_s\| \sum_{i \in \mathcal{X}} \delta \|\phi(\theta_i)\| \\ & \quad +\frac{2\beta}{nR} \|p^*(t) - p_s\| \sum_{i \in \mathcal{Y}} \delta \|\phi(\theta_i)\| \\ & = \frac{2\beta\delta}{R} \|p^*(t) - p_s\|. \end{aligned} \quad (19)$$

Let p_A and p_B be the intersection points of the circle centered at p_s with radius $D_{\max}$ and the circle centered at p^* with radius R . Let ω be the angle of the vector $(p^* - p_s)$. Let φ

be the angle of the vector $(p_A - p^*)$. Since $p_i = p^* + R\phi(\theta_i)$, $D_{\max}^2 = \|p^* - p_s\|^2 + R^2 - 2R\|p^* - p_s\| \cos(\pi - (\varphi - \omega))$, and $\|p_i - p_s\|^2 = \|p^* - p_s\|^2 + R^2 - 2R\|p^* - p_s\| \cos(\pi - (\theta_i - \omega))$, it follows that:

$$\begin{aligned} & \kappa \sum_{i \in \mathcal{Y}} \left(\|p_i - p_s\|^2 - D_{\max}^2 \right) \phi(\theta_i) \\ &= \kappa \sum_{i \in \mathcal{Y}} \left(\|p^* - p_s\|^2 + R^2 - D_{\max}^2 \right) \phi(\theta_i) \\ &\quad + \kappa \sum_{i \in \mathcal{Y}} 2R\|p^* - p_s\| \cos(\theta_i - \omega) \phi(\theta_i) \\ &= \kappa \sum_{i \in \mathcal{Y}} 2R\|p^* - p_s\| \cos(\pi - (\varphi - \omega)) \phi(\theta_i) \\ &\quad + \kappa \sum_{i \in \mathcal{Y}} 2R\|p^* - p_s\| \cos(\theta_i - \omega) \phi(\theta_i) \\ &= 2\kappa R\|p^* - p_s\| \sum_{i \in \mathcal{Y}} \cos(\theta_i - \omega) \phi(\theta_i) \\ &\quad - 2\kappa R\|p^* - p_s\| \sum_{i \in \mathcal{Y}} \cos(\varphi - \omega) \phi(\theta_i). \end{aligned}$$

Without the loss of generality, let $\omega = 0$. It follows that:

$$\begin{aligned} & \frac{2\beta\kappa}{nR} \sum_{i \in \mathcal{Y}} \left(\|p_i - p_s\|^2 - D_{\max}^2 \right) \phi(\theta_i) \\ &= \frac{4\beta\kappa}{n} \|p^*(t) - p_s\|^2 \left(\sum_{i \in \mathcal{Y}} \cos^2 \theta_i - \cos \varphi \sum_{i \in \mathcal{Y}} \cos \theta_i \right). \end{aligned}$$

Since $\sum_{i=1}^d (2\pi/n) \cos(2\pi i/n) \leq \sin(2\pi d/n)$ for all $d \in (1, n)$, one has $\sum_{i=1}^d \cos(2\pi i/n) \leq n \sin(2\pi d/n)/2\pi$ and $\sum_{i=1}^d \cos(4\pi i/n) \leq n \sin(4\pi d/n)/4\pi$. By trigonometric properties, the following holds: $\cos^2 \theta_i = (\cos 2\theta_i + 1)/2$, $\sum_{i \in \mathcal{Y}} \cos^2 \theta_i = n_Y/2 + (1/2) \sum_{i \in \mathcal{Y}} \cos 2\theta_i$. Consequently, $\sum_{i \in \mathcal{Y}} \cos \theta_i \leq n \sin(\pi n_Y/n)/\pi$ and $\sum_{i \in \mathcal{Y}} \cos^2 \theta_i \leq n_Y/2 + n \sin(2\pi n_Y/n)/4\pi$. Since $\varphi/\pi = n_Y/n$, one has $\cos \varphi \geq 0$ and $\cos \varphi < 0$, when $0 \leq n_Y \leq n/2$ and $n/2 < n_Y \leq n - 1$. When $1 \leq n_X < n/2$, one has

$$\begin{aligned} & \frac{2\beta\kappa}{nR} \sum_{i \in \mathcal{Y}} \left(\|p_i - p_s\|^2 - D_{\max}^2 \right) \phi(\theta_i) \\ &= \frac{4\beta\kappa}{n} \|p^*(t) - p_s\|^2 \left(\sum_{i \in \mathcal{Y}} \cos^2 \theta_i - \cos \varphi \sum_{i \in \mathcal{Y}} \cos \theta_i \right) \\ &\leq 2\beta\kappa \left(\frac{n_Y}{n} + \frac{\sin(\frac{2\pi n_Y}{n})}{2\pi} \right) \|p^*(t) - p_s\|^2 \\ &\quad - \frac{4\beta\kappa}{n} \left(\frac{n}{\pi} \cos\left(\frac{\pi n_Y}{n}\right) \sin\left(\frac{\pi n_Y}{n}\right) \right) \|p^*(t) - p_s\|^2 \\ &= 2\beta\kappa \left(\frac{n_Y}{n} + \frac{\sin(\frac{2\pi n_Y}{n})}{2\pi} \right) \|p^*(t) - p_s\|^2 \\ &\quad - 2\beta\kappa \left(\frac{\sin(\frac{2\pi n_Y}{n})}{\pi} \right) \|p^*(t) - p_s\|^2 \end{aligned}$$

$$= 2\beta\kappa \left(\frac{n\gamma}{n} - \frac{\sin(\frac{2\pi n\gamma}{n})}{2\pi} \right) \|p^*(t) - p_s\|^2. \quad (20)$$

It follows from (18) to (20) that:

$$\begin{aligned} \dot{V}(p^*(t)) &\leq -2\beta\kappa(p^*(t) - p_s)^T(p^*(t) - p_s) \\ &\quad + \frac{2\beta\delta}{R} \|p^*(t) - p_s\| \\ &\quad + 2\beta\kappa \left(\frac{n\gamma}{n} - \frac{\sin(\frac{2\pi n\gamma}{n})}{2\pi} \right) \|p^*(t) - p_s\|^2 \\ &= -2\beta\kappa \left(\frac{n\chi}{n} - \frac{\sin(\frac{2\pi n\chi}{n})}{2\pi} \right) \|p^*(t) - p_s\|^2 \\ &\quad + \frac{2\beta\delta}{R} \|p^*(t) - p_s\| \\ &= -\frac{\beta\kappa(2\pi n\chi - n \sin(\frac{2\pi n\chi}{n}))}{n\pi} \|p^*(t) - p_s\|^2 \\ &\quad + \frac{2\beta\delta}{R} \|p^*(t) - p_s\| \\ &= -\frac{\beta\kappa(2\pi n\chi - n \sin(\frac{2\pi n\chi}{n}))}{n\pi} \|p^*(t) - p_s\| \\ &\quad \times \left(\|p^*(t) - p_s\| - \frac{2\pi n\delta}{\kappa R(2\pi n\chi - n \sin(\frac{2\pi n\chi}{n}))} \right). \end{aligned}$$

When $n/2 \leq n_\chi \leq n$, it follows that:

$$\begin{aligned} &\frac{2\beta\kappa}{nR} \sum_{i \in \mathcal{Y}} \left(\|p_i - p_s\|^2 - D_{\max}^2 \right) \phi(\theta_i) \\ &= \frac{4\beta\kappa}{n} \|p^*(t) - p_s\|^2 \sum_{i \in \mathcal{Y}} \cos^2 \theta_i \\ &\leq 2\beta\kappa \left(\frac{n\gamma}{n} + \frac{\sin(\frac{2\pi n\gamma}{n})}{2\pi} \right) \|p^*(t) - p_s\|^2. \quad (21) \end{aligned}$$

Similarly, when $t > T_c$ and $n/2 \leq n_\chi \leq n$, one has

$$\begin{aligned} \dot{V}(p^*(t)) &\leq -2\beta\kappa(p^*(t) - p_s)^T(p^*(t) - p_s) \\ &\quad + \frac{2\beta\delta n_\chi}{nR} \|p^*(t) - p_s\| \\ &\quad + 2\beta\kappa \left(\frac{n\gamma}{n} + \frac{\sin(\frac{2\pi n\gamma}{n})}{2\pi} \right) \|p^*(t) - p_s\|^2 \\ &= -\frac{\beta\kappa(2\pi n_\chi + n \sin(\frac{2\pi n_\chi}{n}))}{n\pi} \|p^*(t) - p_s\| \\ &\quad \times \left(\|p^*(t) - p_s\| - \frac{2\pi n\delta}{\kappa R(2\pi n_\chi + n \sin(\frac{2\pi n_\chi}{n}))} \right). \end{aligned}$$

Let $G_1 = \frac{2\pi n\delta}{\kappa R(2\pi n_\chi - n \sin(\frac{2\pi n_\chi}{n}))}$, $G_2 = \frac{2\pi n\delta}{\kappa R(2\pi n_\chi + n \sin(\frac{2\pi n_\chi}{n}))}$, and $G = \frac{2\pi n\delta}{\kappa R(2\pi n_\chi - n \sin(\frac{2\pi n_\chi}{n}))}$. One has $G \geq G_1$ and $G \geq G_2$.

Thus, $\dot{V}(p^*(t)) < 0$ for all $n_\chi \in [1, n]$, when $\|p^*(t) - p_s\| > G$. Let Ω_p be the closed ball centered at p_s with radius G . Let

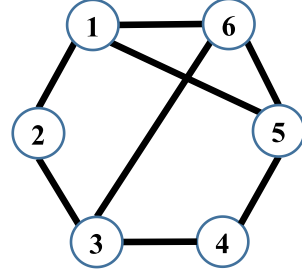


Fig. 1. Network topology of the robots in simulation.

$\Omega_l = \{p^* : V(p^*) \leq l\}$ be the level set of the Lyapunov function $V(p^*)$ with $l = G^2/2$. It follows that $V(p^*) = \frac{1}{2}\|p^* - p_s\|^2 \leq G^2/2 = l$ when $\|p^* - p_s\| \leq G$, i.e., $\Omega_p \in \Omega_l$. As a result, any solution of (17) starting in $\mathbb{R}^2/\Omega_l$ satisfies $\dot{V}(p^*) < 0$. Thus, it will enter Ω_l in finite time and remains in it thereafter. This guarantees the asymptotic stability of $p^* = p_s$ in the ball with a radius ϵ , and the solutions will enter the region $\|p^* - p_s\| \leq G$, where

$$\epsilon = \frac{2\pi n\delta}{\kappa R(2\pi n_\chi - n \sin(\frac{2\pi n_\chi}{n}))}.$$

This implies that $\|p^* - p_s\| \leq \epsilon$ as $t \rightarrow \infty$, which gives $\lim_{t \rightarrow \infty} \|p^*(t) - p_s\| \leq \epsilon$ and $\lim_{t \rightarrow \infty} \|p_i(t) - R\phi(\theta_i) - p_s\| \leq \epsilon$ for all $i \in \mathcal{V}$. ■

Remark 4: It follows from Theorem 1 that, when the number of informed robots $n_\chi = n$, the localization error is further reduced to $\epsilon = \delta/\kappa R$.

Remark 5: When the disturbance intensity δ decreases, the localization error ϵ will be reduced. Particularly, as $\delta \rightarrow 0$, one has $\epsilon \rightarrow 0$.

Remark 6: In Theorem 1, the formation control gain α needs to be sufficiently large to satisfy the condition $\alpha/\beta > 4nf_{D_{\max}}/R$.

Remark 7: The above results can be extended to the higher dimensional case by employing the technique of balanced formations [23]. The approach can also be adopted to handle more complex robot dynamics, e.g., unicycle model, with the aid of feedback linearization (see Section V for details).

IV. SIMULATION

This section presents a numerical example of six mobile robots, with the network topology, as illustrated in Fig. 1. The signal distribution function is $f(z) = \|z - p_s\|^2$, where $p_s = [5.5, 5.5]^T$. The measurements of the robots are corrupted by Gaussian noise $\eta(p_i) \sim N(0, 0.2)$. The other parameters are chosen as $R = 2$ and $D_{\max} = 12$.

In Fig. 2, the circles represent different robots, while the dashed lines represent the trajectories of the robots. It can be observed that under the proposed algorithm, all robots achieve a circular formation, and finally, the formation center is steered into a neighborhood of the source. Fig. 3 demonstrates the formation error of the robots, which shows that the formation error vanishes asymptotically. Fig. 4 shows the localization error of the robots. It can be seen that the localization error converges

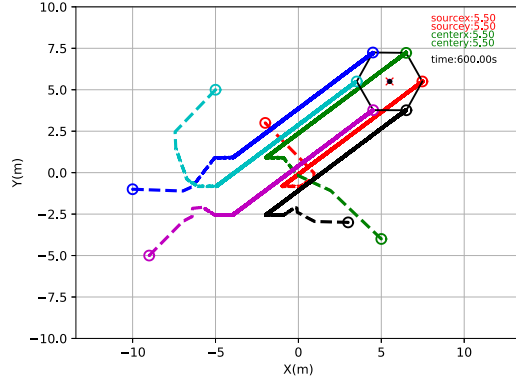


Fig. 2. Trajectories of the robots in simulation.

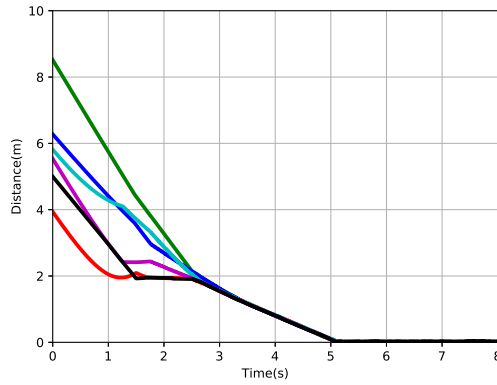


Fig. 3. Formation error of the robots in simulation.

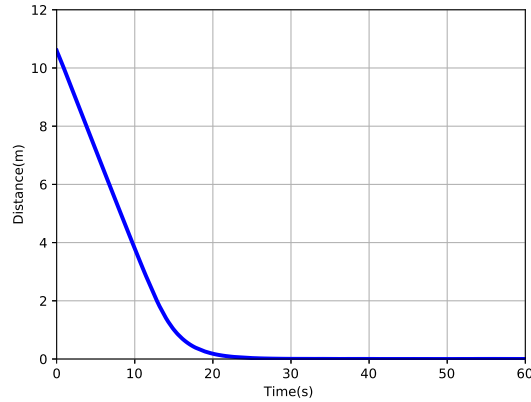


Fig. 4. Localization error of the robots in simulation.

to a small value asymptotically. It is noted also from Figs. 3 and 4 that formation is achieved earlier than localization, which is consistent with the analysis in the proof of Theorem 1.

V. EXPERIMENTS

In this section, experiments are performed to demonstrate the effectiveness of the control law (4). The system consists of an Optitrack system and a group of TurtleBot3 robots (see Fig. 5). The kinematics of robot i is described by

$$\dot{x}_i = v_i \cos \theta_i$$

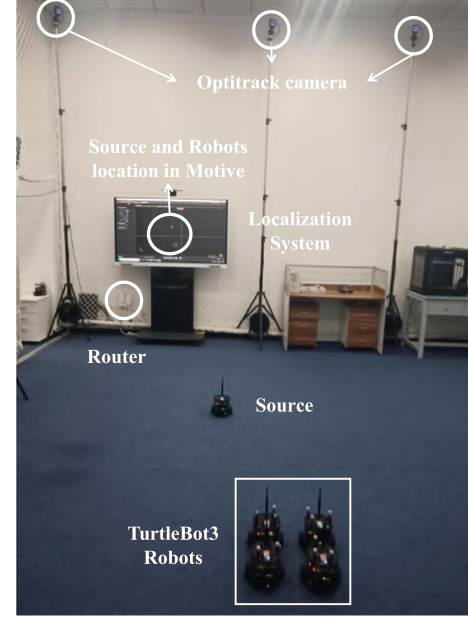


Fig. 5. Experimental platform.

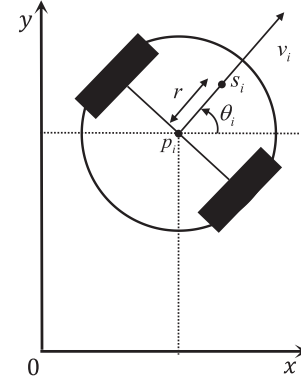


Fig. 6. Unicycle model.

$$\dot{y}_i = v_i \sin \theta_i$$

$$\dot{\theta}_i = w_i$$

where $p_i = [x_i, y_i]^T$ and θ_i denote, respectively, the position and orientation, and v_i and w_i are the linear and angular velocity of robot i (see Figs. 6 and 7). Let $s_i = [x_{si}, y_{si}]^T$ be a point shifted from the center p_i by r , i.e.,

$$x_{si} = x_i + r \cos \theta_i$$

$$y_{si} = y_i + r \sin \theta_i.$$

The kinematics of the point s_i is given by

$$\begin{bmatrix} \dot{x}_{si} \\ \dot{y}_{si} \end{bmatrix} = \begin{bmatrix} \cos \theta_i & -r \sin \theta_i \\ \sin \theta_i & r \cos \theta_i \end{bmatrix} \begin{bmatrix} v_i \\ w_i \end{bmatrix} = f_i$$

where f_i is defined by the control law (4). Once f_i is designed, the actual control inputs v_i and w_i can be recovered from f_i as

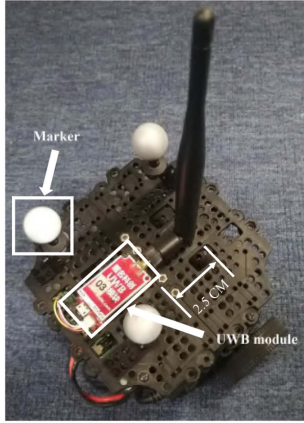


Fig. 7. TurtleBot3 robot.

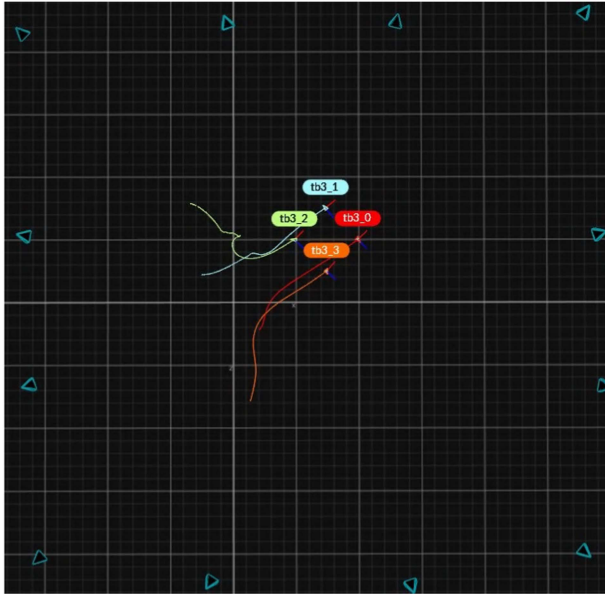


Fig. 8. Trajectories of the robots in the experiment.

follows:

$$\begin{aligned} \begin{bmatrix} v_i \\ \omega_i \end{bmatrix} &= \begin{bmatrix} \cos \theta_i & -r \sin \theta_i \\ \sin \theta_i & r \cos \theta_i \end{bmatrix}^{-1} f_i \\ &= \begin{bmatrix} \cos \theta_i & \sin \theta_i \\ -\frac{1}{r} \sin \theta_i & \frac{1}{r} \cos \theta_i \end{bmatrix} f_i. \end{aligned}$$

In the experiments, the parameters are set as $R = 0.5$ m and $r = 2.5$ cm. Each robot's measurement is generated by the Optitrack system, which is corrupted Gaussian noise $\eta(p_i) \sim N(0, 0.2)$. The real-time trajectories of the TurtleBot3 robots are given by the Optitrack system (see Fig. 8). The communication is realized in a sampled-data manner. That is, after every $T = 0.1$ s, a robot will send its information to all neighboring robots. Fig. 9 shows the formation error, while Fig. 10 demonstrates the localization error. It can be seen from Figs. 9 and 10 that, under the proposed control law, the circular formation is attained and the source is localized.

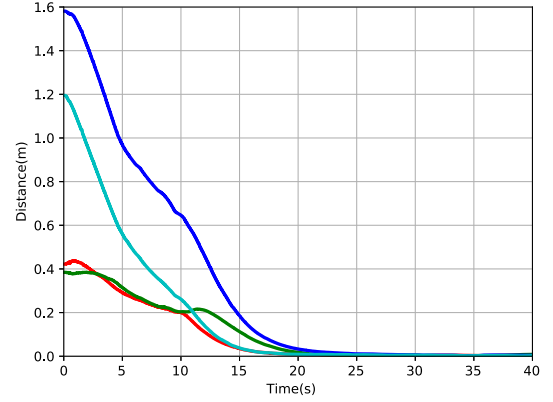


Fig. 9. Formation error of the robots in the experiment.

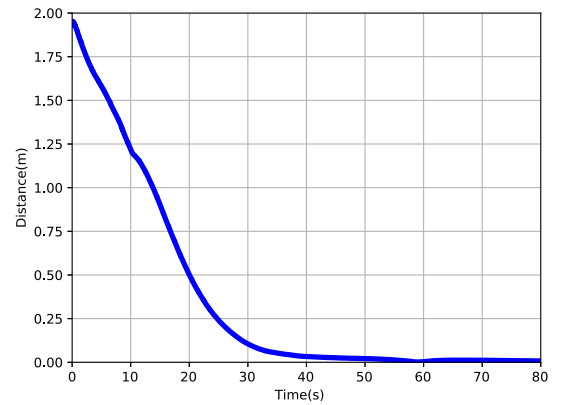


Fig. 10. Localization error of the robots in the experiment.

VI. CONCLUSION

This article has investigated simultaneous source localization and formation control with noisy measurements and ternary communication. The primary results are twofold.

- 1) A novel sign gradient-free algorithm is proposed, which is capable of handling the measurement noise and reducing the communication cost.
- 2) An inherent connection is established between the selection of control parameters and the convergence property of the resulting algorithm. It is demonstrated that the localization error can be made arbitrarily small by tuning the control parameter, along with signal strength, noisy intensity, formation radius, and the number of informed robots.

Future works include possible extensions to multirobot systems with time-varying sources, directed network topologies, and delayed and unsafe communications.

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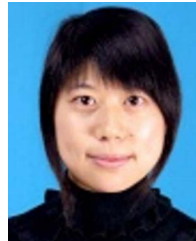


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